jigsaw-agile-community-rules

13th Place Solution: Using Positive Examples from Other Rules as Negative Exampl

Rank	13
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Source: https://www.kaggle.com/competitions/jigsaw-agile-community-rules/discussion/613265
Retrieved with Kaggle CLI on 2026-10-21.

First of all, I would like to thank the competition organizers for hosting this event, which provided us with a valuable opportunity to learn and explore new techniques.
Next, I would like to express my sincere gratitude to my supervisor Professor @zwzwyang , as well as my two teammates @mavicbf and @mikeoscar1 — each of them played an indispensable role throughout the process.
Finally, I would like to thank the Guangdong Smart Education Research Institute of Jinan University for its support.

Method Overview:
Our main idea is to use the positive and negative samples from the public test set (Public) for training.
During data augmentation, we convert positive examples under other rules into negative examples under the current rule (30%)(Public-1) to better fit the test distribution.

```
# Sample 30% of the data
sampled = dataframe.sample(frac=0.3, random_state=42).copy()

all_rules = dataframe["rule"].unique().tolist()
def get_different_rule(current_rule):
choices = [r for r in all_rules if r != current_rule]
return pd.Series(choices).sample(1).iloc[0]

sampled["rule"] = sampled["rule"].apply(get_different_rule)
sampled["rule_violation"] = 0 # Always negative
dataframe = pd.concat([dataframe, sampled], axis=0).reset_index(drop=True)
```

Model Ensemble:

Model	Public Score	Public-1 Score	Private Score	Training Time
Qwen3-8b-gptq-int4	0.922	0.927	0.923	8h
Qwen3-1.7b-gptq-int4	0.911	0.918	0.912	1.5h
DeBERTa-base	0.906	0.911	0.903	1h
DeBERTa-small	0.900	–	–	0.5h

We did not use additional data for DeBERTa-small, as experiments showed that including it slightly degraded the overall ensemble performance.
Our open-source code is available at:
https://www.kaggle.com/code/yangjiahua/finall-qwen3-8b-int4

Additional Ideas:
1.Pretraining on Test Body Text. Before training DeBERTa, we could pretrain it using the body column from test.csv, which might be helpful. Due to time constraints, we didn't manage to explore this idea.
2.Reinforcement Learning (RL) Fine-tuning. I experimented with reinforcement learning on Qwen3-1.7B, which performed better than LoRA fine-tuning. With a high-quality dataset and RL training, we believe performance could be further improved.
3.Dataset Construction with LLMs. We used large language models (such as Gemini and Qwen) to generate training data for private rules in the test set.
For example, the synthetic dataset for Private Rule 3 helped improve performance — boosting Qwen2.5-14B-Instruct from 0.886 → 0.888. Unfortunately, other generated rules had a negative effect on model performance.

Supplemental Q&A

CPMP · 2025-10-25T19:36:40.490000

Congrats on the result.

> Our main idea is to use the positive and negative samples from the public test set (Public) for training.

Do you know how much you go from using it compared ot not using it?

I noticed it was almost perfect on train data but did not had time to submit it. I wasn't sure it would help as it increases the training data a lot. hence we can use less models.

YANGJIAHUA · 2025-10-26T13:02:59.310000

Hello, if we train Qwen3 solely on train.csv without using any information from test.csv, its performance is worse than Qwen2.5. Using only train.csv, our Qwen2.5-14B-Instruct model achieved a score of 0.886.

CPMP · 2025-10-26T19:09:48.100000

Hi, thanks but I want to know something different. I wasn't clear enough.

I found, like you, that positive examples for one rule are negative examples for the other rules (almost always).

I did not use that. I trained using the positive and negative examples, rule by rule.

My question is:
did you compare with and without using positive form other rules as negatives?

YANGJIAHUA · 2025-10-27T06:49:54.163000

Our Public Score refers to the score without using positive examples from other rules as negatives, while the Public-1 Score refers to the score with using positive examples from other rules as negatives.

Absinthe · 2026-03-30T16:03:43.013000

May I ask about the RL-based fine-tuning mentioned above. Is there any public code available to see for 'I conducted reinforcement learning experiments on the Qwen3-1.7B dataset, and its performance is better than LoRA fine-tuning'? Thank you.

YANGJIAHUA · 2026-03-31T10:22:00.480000

Hello, for this part of the code, you can implement it using some mainstream RL training frameworks, such as OpenR1 or VERL.

Absinthe · 2026-04-01T09:27:08.140000

Thank you, may I ask if you have tried reinforcement learning methods? Did you use DPO or something else? I saw you mentioned that the effect of using reinforcement learning methods is better than SFT. Could you tell me about it? I am very interested in this, thank you.

YANGJIAHUA · 2026-04-01T10:04:44.907000

> Hello! Thank you very much for your question. I am using my own modified version of GRPO. Since I am currently still writing the paper, I do not plan to release the related code for now. You can directly use VERL's GRPO to achieve good performance. We plan to release our modified GRPO method in May; it can achieve a 2× improvement in training speed and also better performance. If you are interested in reinforcement learning, I will share our related paper with you later. Thank you!